

Inference

Grad admissions data

Interested in predicting grad admission from GPA, GRE score, and school rank:

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Suppose we want to estimate the effect of GRE, after accounting for GPA and school rank. How would I compute a confidence interval for β_2 ?

Previously: confidence intervals

Normal confidence interval for β_2 :

↑
using normal
distribution

$$\hat{\beta}_2 \pm z^* SE_{\hat{\beta}_2}$$

critical value from $N(0,1)$
e.g. $z^* = 1.96$ for 95%
 $z^* = 1.645$ for 90%

Options for estimating $SE_{\hat{\beta}_2}$:

- + Bootstrap estimate of standard error
- + Estimate of standard error from R output

Confidence interval

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.989979073	1.139950936	-3.500132	0.0004650273
gpa	0.804037549	0.331819298	2.423119	0.0153878974
gre	0.002264426	0.001093998	2.069864	0.0384651284
as.factor(rank)2	-0.675442928	0.316489661	-2.134171	0.0328288188
as.factor(rank)3	-1.340203916	0.345306418	-3.881202	0.0001039415
as.factor(rank)4	-1.551463677	0.417831633	-3.713131	0.0002047107

95% CI for β_2 : $0.00226 \pm 1.96 (0.00109)$
 $= (0.00012, 0.0044)$

95% CI for e^{β_2} : $(e^{0.00012}, e^{0.0044}) = (1.00012, 1.0044)$

How do I interpret this confidence interval?

We are 95% confident that a one-point increase in GRE score is associated with an increase in odds of acceptance by between 0.012% and 0.44%, holding GPA & school rank fixed

Another question

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

How could we address this research question?

Hypothesis testing

What steps are involved in hypothesis testing?

- choose desired type I error rate α
- specify H_0 & H_A
- compute a test statistic
- P-value : how unusual is the observed test statistic, if H_0 is true?
- Null distribution : distribution of test stat. under H_0

Specifying the hypotheses

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

$H_0:$ $\beta_2 = 0$ (no relationship)

$H_A:$ $\beta_2 \neq 0$ (some relationship)

Confidence intervals and hypothesis testing

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

$$H_0 : \beta_2 = 0 \quad H_A : \beta_2 \neq 0$$

Recall that we would **reject** H_0 at significance level α if and only if a $(1 - \alpha)100\%$ confidence interval for β_2 **does not** contain 0. (Equivalently, a $(1 - \alpha)100$ confidence interval for e^{β_2} **does not** contain 1).

$$95\% \text{ CI for } \beta_2 : (0.00012, 0.0044)$$

$$95\% \text{ CI for } e^{\beta_2} : (1.00012, 1.0044)$$

$$\underline{\text{reject}} H_0 : \beta_2 = 0 \quad \text{at level } \alpha = 0.05$$

Computing a p-value

linear: $y = \beta_0 + \beta_1 x + \varepsilon$
 $\varepsilon \sim N(0, \sigma^2)$
 $\uparrow$
need to estimate!

To compute a p-value, we need:

logistic regression: no σ^2 to estimate

- + a test statistic
- + the distribution of the test statistic if H_0 were correct

What test statistic did we use for testing single coefficients in *linear* models?

$$H_0: \beta_j = b$$

statistic:
$$\frac{\hat{\beta}_j - b}{SE \hat{\beta}_j}$$

same idea for logistic regression!

$$H_0: \beta_2 = 0 \quad Z = \frac{\hat{\beta}_2 - 0}{SE \hat{\beta}_2} = 2.07$$

The null distribution

To compute a p-value, we need the **null distribution** (distribution of the test statistic under H_0).

Does *bootstrap resampling* (what we used to create a confidence interval) give us the **null** distribution?

Not quite !

Bootstrap resampling mimics sampling
from population
we don't know if H_0 is true in
population !

Parametric bootstrapping

if H_0 is true: don't need GRE!

→ Reduced model: $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 RankZ_i + \beta_3 RankB_i + \beta_4 RankU_i$

① Fit reduced model

② Simulate response from reduced model (H_0 guaranteed true!) $Admit_i^* \sim \text{Bernoulli}(\hat{\pi}_{i, \text{reduced}})$

③ Refit full model (GRE included!) on simulated data → $\hat{\beta}_{\text{sim}}$

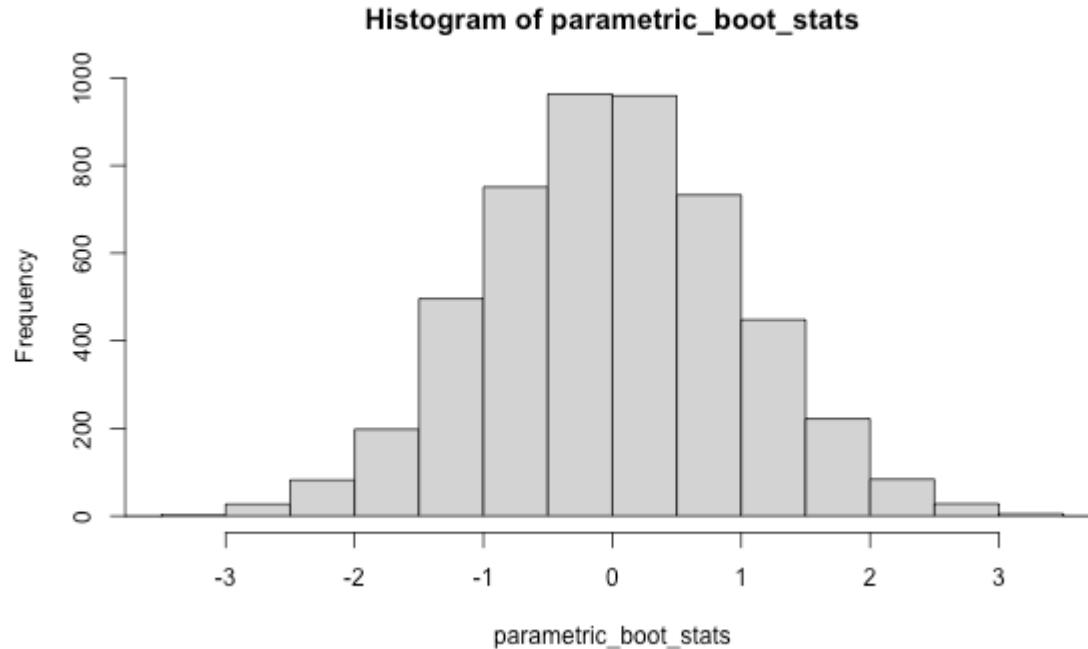
④ Recalculate test statistic: $Z_{\text{sim}} = \frac{\hat{\beta}_{\text{sim}}}{SE \hat{\beta}_{\text{sim}}}$

⑤ Repeat many times!

Class activity

- + Work on the class activity (linked on Canvas and course website)
- + Submit rendered HTML on Canvas at the end of class

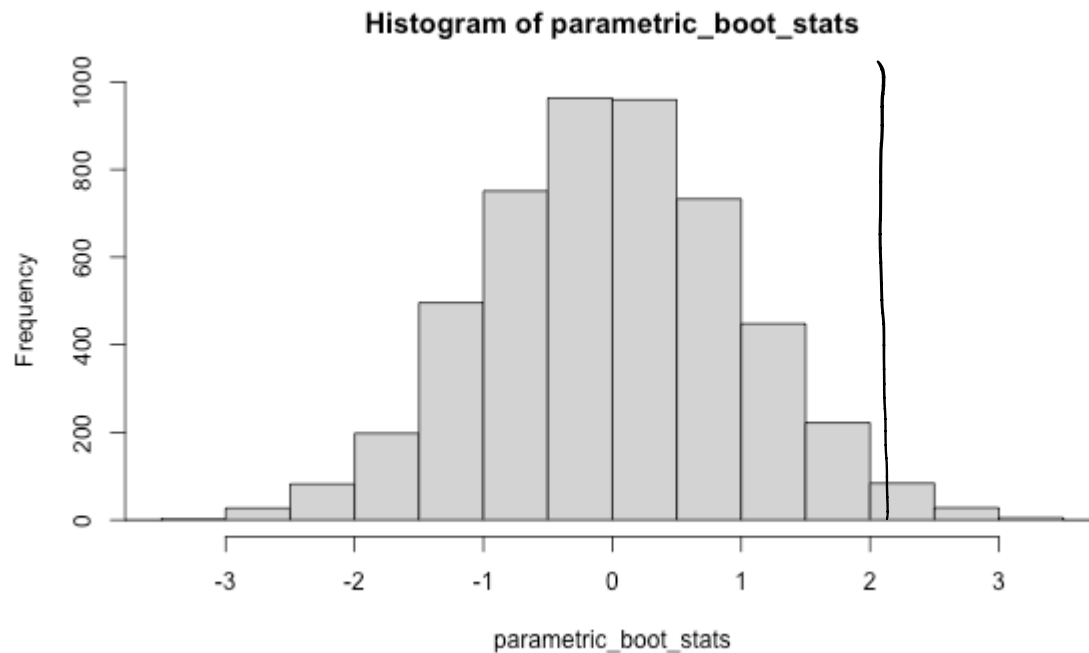
Activity



What does this distribution look like?

a $N(0,1)$ distribution!

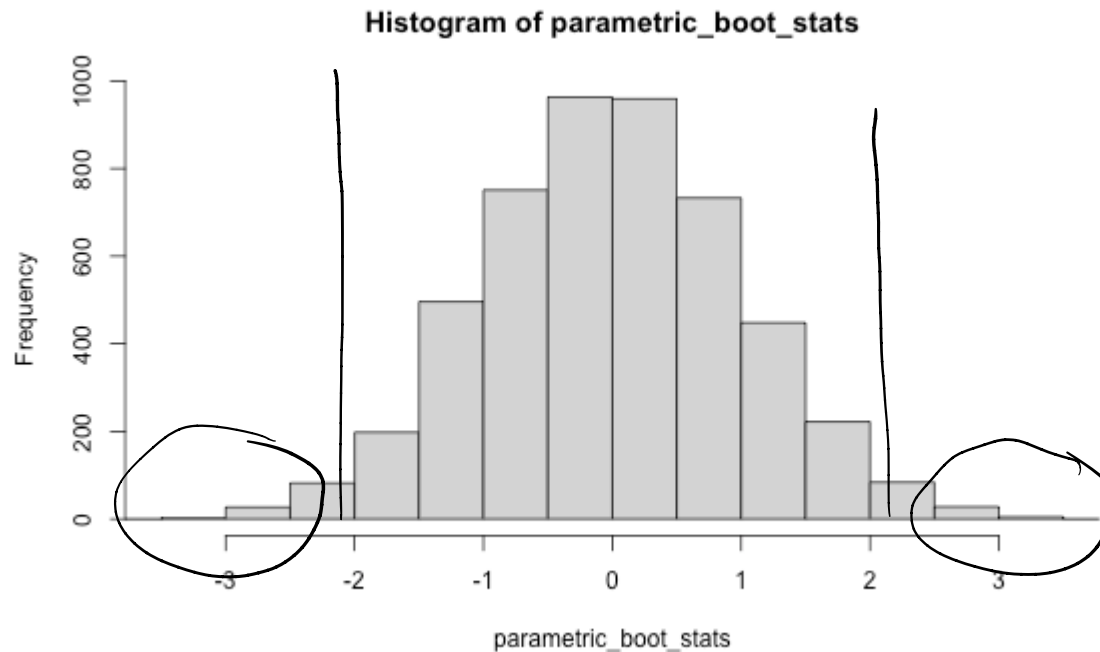
Activity



How unusual is the observed test statistic of 2.07, compared to this null distribution?

$$H_A: \beta_2 \neq 0$$

p-value



```
mean(abs(parametric_boot_stats) > 2.07)
```

```
## [1] 0.0394
```

p-value

```
mean(abs(parametric_boot_stats) > 2.07)
```

```
## [1] 0.0394
```

```
summary(m_full)$coefficients
```

##		Estimate	Std. Error	z value	Pr(> z)
##	(Intercept)	-3.989979073	1.139950936	-3.500132	0.0004650273
##	gpa	0.804037549	0.331819298	2.423119	0.0153878974
##	gre	0.002264426	0.001093998	2.069864	0.0384651284
##	as.factor(rank)2	-0.675442928	0.316489661	-2.134171	0.0328288188
##	as.factor(rank)3	-1.340203916	0.345306418	-3.881202	0.0001039415
##	as.factor(rank)4	-1.551463677	0.417831633	-3.713131	0.0002047107

↑
p-values based on
 $N(0,1)$ for
testing $H_A: \beta_j \neq 0$